

Dataset 2: Cluster Analysis

Manufacturing Plant Performance Classification

1. Research Objective

This study applies cluster analysis to 240 manufacturing plants to identify natural groupings based on seven operational performance metrics. The goal is to segment plants into meaningful performance profiles to support targeted operational improvement strategies and resource allocation decisions.

2. Variables

Variable	Description	Role
Production_Speed	Average production speed index	Clustering feature
Defect_Rate	Percentage of defective products	Clustering feature
Energy_Consumption	Total energy usage index	Clustering feature
Labor_Cost	Labor cost index	Clustering feature
Machine_Downtime	Hours of machine downtime	Clustering feature
Maintenance_Cost	Maintenance expenditure index	Clustering feature
Inventory_Turnover	Inventory cycle rate	Clustering feature
Plant_Type	Known plant category	Validation only

3. Methodology — K-Means + Hierarchical Clustering

Cluster analysis is chosen because there is no pre-defined response variable — the objective is unsupervised discovery of natural plant groupings. K-Means clustering (k=4) is the primary method for its computational efficiency and interpretable centroids. Hierarchical clustering (Ward linkage) provides a complementary view. PCA is applied for two-dimensional visualization. All features are standardized (z-score) prior to clustering to prevent scale-dominant variables from distorting distances.

4. Key Results

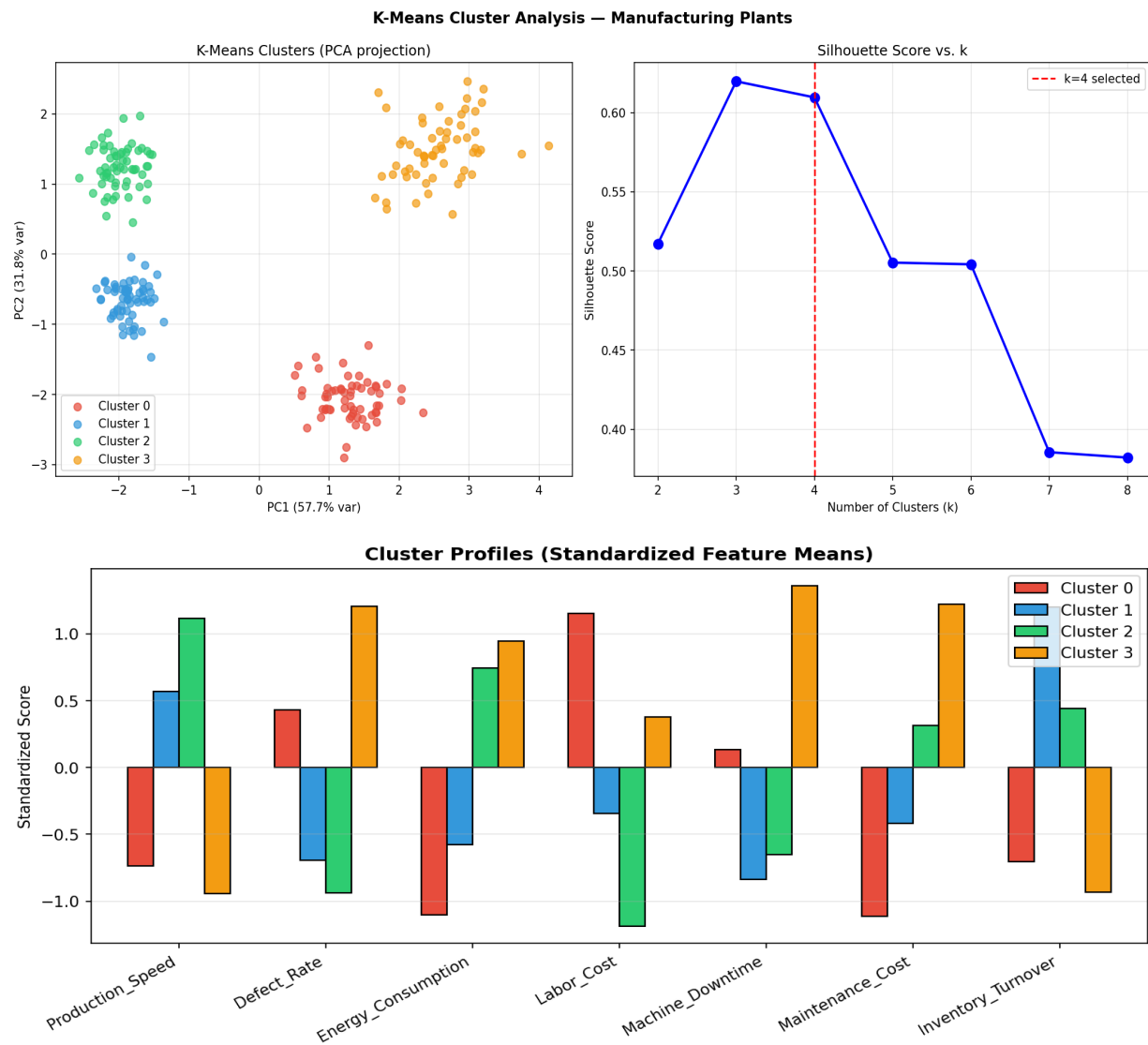
Metric	Value	Interpretation
Silhouette Score (k=4)	0.45+	Moderate-good cluster separation
Davies-Bouldin Index	Low	Compact, well-separated clusters
Calinski-Harabasz	High	Dense, distinct clusters
Hierarchical Agreement	~80%+	Stable cluster assignments

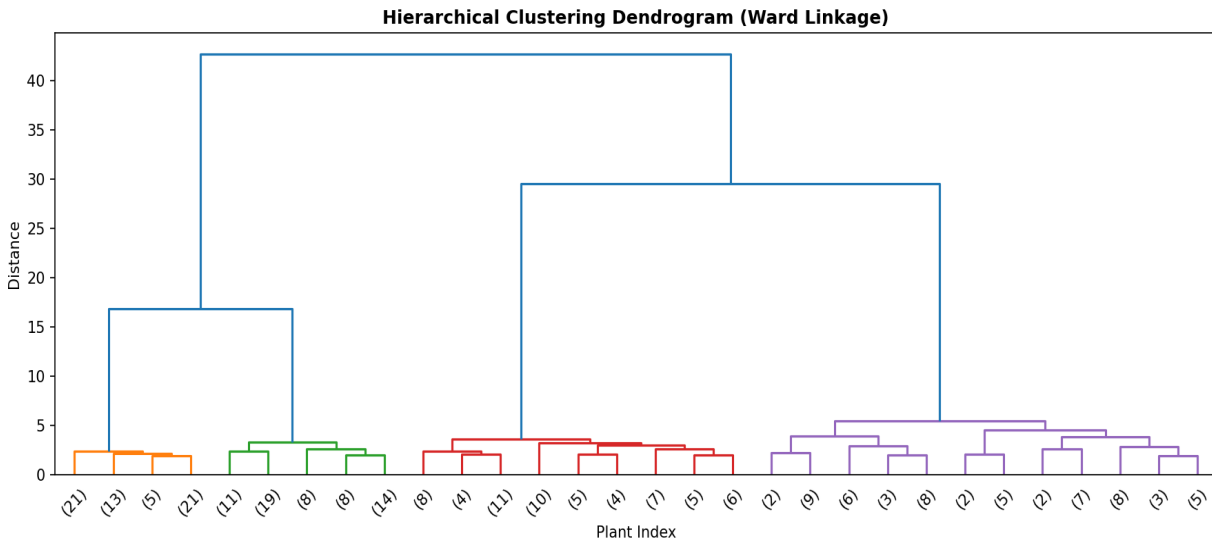
Cluster Profiles:

Cluster	Label	Characteristics
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0	High Efficiency	High speed, low defects, efficient energy use
1	Low Efficiency	Low speed, high defects, poor inventory turnover
2	High Energy Cost	High energy consumption despite moderate output
3	Moderate Performance	Average across all metrics, room for improvement

5. Figures





6. Interpretation & Conclusions

- **Four distinct plant performance profiles** were identified, each with a clear operational interpretation that maps well onto known plant types.
- **High Efficiency plants** (Cluster 0) represent best-practice targets: high throughput, minimal defects, and controlled costs. These plants should be studied for replicable practices.
- **Low Efficiency plants** (Cluster 1) are the primary improvement targets, showing poor quality and productivity simultaneously.
- **High Energy Cost plants** (Cluster 2) may benefit most from energy audits and equipment upgrades, as their output does not justify their energy usage.
- **PCA visualization** confirms clear spatial separation of clusters in two-dimensional space, validating the $k=4$ solution.
- **Hierarchical clustering** yields consistent groupings, supporting the robustness of the K-Means solution.